110. **[T]** A professor asks her class to report the amount of time t they spent writing two assignments. Most students report that it takes them about 45 minutes to type a four-page assignment and about 1.5 hours to type a nine-page assignment.
   a. Find the linear function $y = N(t)$ that models this situation, where N is the number of pages typed and t is the time in minutes.
   b. Use part a. to determine how many pages can be typed in 2 hours.
   c. Use part a. to determine how long it takes to type a 20-page assignment.

111. **[T]** The output (as a percent of total capacity) of nuclear power plants in the United States can be modeled by the function $P(t) = 1.8576t + 68.052$, where t is time in years and $t = 0$ corresponds to the beginning of 2000. Use the model to predict the approximate percentage output at the beginning of 2015.

112. **[T]** The admissions office at a public university estimates that 65% of the students offered admission to the class of 2019 will actually enroll.
   a. Find the linear function $y = N(x)$, where N is the number of students that actually enroll and x is the number of all students offered admission to the class of 2019.
   b. If the university wants the 2019 freshman class size to be 1350, determine how many students should be offered admission.

1.3 Trigonometric Functions

Learning Objectives

1.3.1 Convert angle measures between degrees and radians.
1.3.2 Recognize the triangular and circular definitions of the basic trigonometric functions.
1.3.3 Write the basic trigonometric identities.
1.3.4 Identify the graphs and periods of the trigonometric functions.
1.3.5 Describe the shift of a sine or cosine graph from the equation of the function.

Trigonometric functions are used to model many phenomena, including sound waves, vibrations of strings, alternating electrical current, and the motion of pendulums. In fact, almost any repetitive, or cyclical, motion can be modeled by some combination of trigonometric functions. In this section, we define the six basic trigonometric functions and look at some of the main identities involving these functions.

Radian Measure

To use trigonometric functions, we first must understand how to measure the angles. Although we can use both radians and degrees, **radians** are a more natural measurement because they are related directly to the unit circle, a circle with radius 1. The radian measure of an angle is defined as follows. Given an angle θ, let s be the length of the corresponding arc on the unit circle (Figure 1.30). We say the angle corresponding to the arc of length 1 has radian measure 1.

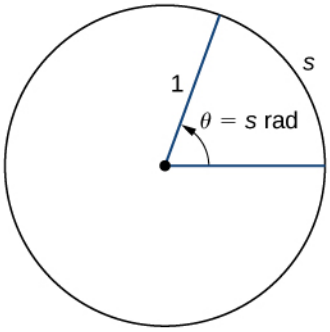


Figure 1.30 The radian measure of an angle θ is the arc length s of the associated arc on the unit circle.

Since an angle of $360°$ corresponds to the circumference of a circle, or an arc of length 2π, we conclude that an angle with a degree measure of $360°$ has a radian measure of 2π. Similarly, we see that $180°$ is equivalent to π radians. Table 1.8 shows the relationship between common degree and radian values.

Degrees	Radians	Degrees	Radians
0	0	120	$2\pi/3$
30	$\pi/6$	135	$3\pi/4$
45	$\pi/4$	150	$5\pi/6$
60	$\pi/3$	180	π
90	$\pi/2$		

Table 1.8 Common Angles Expressed in Degrees and Radians

EXAMPLE 1.22

Converting between Radians and Degrees

a. Express $225°$ using radians.
b. Express $5\pi/3$ rad using degrees.

Solution

Use the fact that $180°$ is equivalent to π radians as a conversion factor: $1 = \frac{\pi \text{ rad}}{180°} = \frac{180°}{\pi \text{ rad}}$.

a. $225° = 225° \cdot \frac{\pi}{180°} = \frac{5\pi}{4}$ rad
b. $\frac{5\pi}{3}$ rad $= \frac{5\pi}{3} \cdot \frac{180°}{\pi} = 300°$

1.17 Express $210°$ using radians. Express $11\pi/6$ rad using degrees.

The Six Basic Trigonometric Functions

Trigonometric functions allow us to use angle measures, in radians or degrees, to find the coordinates of a point on any circle—not only on a unit circle—or to find an angle given a point on a circle. They also define the relationship among the sides and angles of a triangle.

To define the trigonometric functions, first consider the unit circle centered at the origin and a point $P = (x, y)$ on the unit circle. Let θ be an angle with an initial side that lies along the positive x-axis and with a terminal side that is the line segment OP. An angle in this position is said to be in *standard position* (Figure 1.31). We can then define the values of the six trigonometric functions for θ in terms of the coordinates x and y.

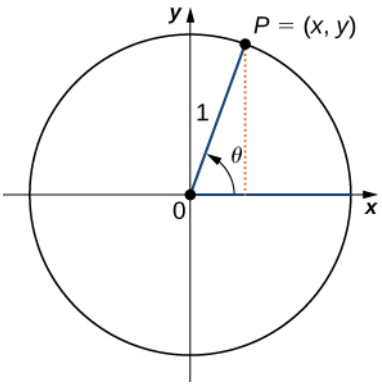


Figure 1.31 The angle θ is in standard position. The values of the trigonometric functions for θ are defined in terms of the coordinates x and y.

Definition

Let $P = (x, y)$ be a point on the unit circle centered at the origin O. Let θ be an angle with an initial side along the positive x-axis and a terminal side given by the line segment OP. The **trigonometric functions** are then defined as

$$\begin{aligned} \sin\theta &= y & \csc\theta &= \frac{1}{y} \\ \cos\theta &= x & \sec\theta &= \frac{1}{x} \\ \tan\theta &= \frac{y}{x} & \cot\theta &= \frac{x}{y} \end{aligned} \tag{1.9}$$

If $x = 0$, $\sec\theta$ and $\tan\theta$ are undefined. If $y = 0$, then $\cot\theta$ and $\csc\theta$ are undefined.

We can see that for a point $P = (x, y)$ on a circle of radius r with a corresponding angle θ, the coordinates x and y satisfy

$$\begin{aligned} \cos\theta &= \frac{x}{r} \\ x &= r\cos\theta \\ \sin\theta &= \frac{y}{r} \\ y &= r\sin\theta. \end{aligned}$$

The values of the other trigonometric functions can be expressed in terms of x, y, and r (Figure 1.32).

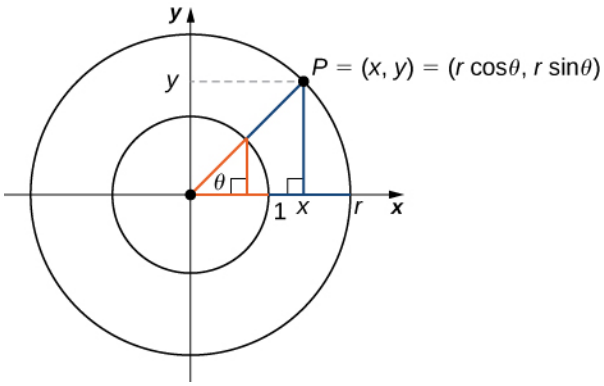


Figure 1.32 For a point $P = (x, y)$ on a circle of radius r, the coordinates x and y satisfy $x = r\cos\theta$ and $y = r\sin\theta$.

Table 1.9 shows the values of sine and cosine at the major angles in the first quadrant. From this table, we can determine the values of sine and cosine at the corresponding angles in the other quadrants. The values of the other trigonometric functions are calculated easily from the values of $\sin\theta$ and $\cos\theta$.

θ	$\sin\theta$	$\cos\theta$
0	0	1
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$
$\frac{\pi}{2}$	1	0

Table 1.9 Values of $\sin\theta$ and $\cos\theta$ at Major Angles θ in the First Quadrant

EXAMPLE 1.23

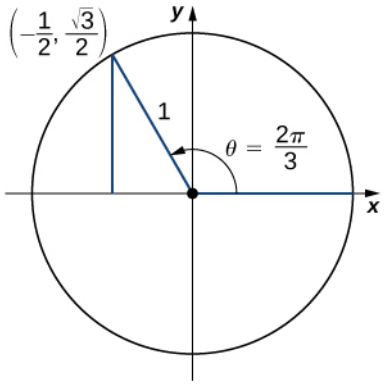
Evaluating Trigonometric Functions

Evaluate each of the following expressions.

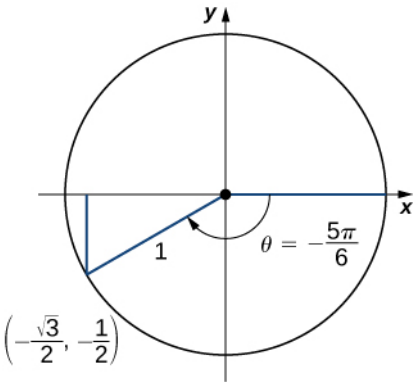
a. $\sin\left(\frac{2\pi}{3}\right)$
b. $\cos\left(-\frac{5\pi}{6}\right)$
c. $\tan\left(\frac{15\pi}{4}\right)$

Solution

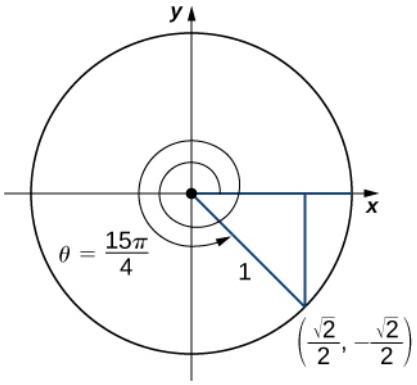
a. On the unit circle, the angle $\theta = \frac{2\pi}{3}$ corresponds to the point $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. Therefore, $\sin\left(\frac{2\pi}{3}\right) = y = \frac{\sqrt{3}}{2}$.



b. An angle $\theta = -\frac{5\pi}{6}$ corresponds to a revolution in the negative direction, as shown. Therefore, $\cos\left(-\frac{5\pi}{6}\right) = x = -\frac{\sqrt{3}}{2}$.



c. An angle $\theta = \frac{15\pi}{4} = 2\pi + \frac{7\pi}{4}$. Therefore, this angle corresponds to more than one revolution, as shown. Knowing the fact that an angle of $\frac{7\pi}{4}$ corresponds to the point $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$, we can conclude that $\tan\left(\frac{15\pi}{4}\right) = \frac{y}{x} = -1$.



☑ 1.18 Evaluate $\cos(3\pi/4)$ and $\sin(-\pi/6)$.

As mentioned earlier, the ratios of the side lengths of a right triangle can be expressed in terms of the trigonometric functions evaluated at either of the acute angles of the triangle. Let θ be one of the acute angles. Let A be the length of the adjacent leg, O be the length of the opposite leg, and H be the length of the hypotenuse. By inscribing the triangle into a circle of radius H, as shown in Figure 1.33, we see that A, H, and O satisfy the following relationships with θ:

$$\sin\theta = \frac{O}{H} \qquad \csc\theta = \frac{H}{O}$$
$$\cos\theta = \frac{A}{H} \qquad \sec\theta = \frac{H}{A}$$
$$\tan\theta = \frac{O}{A} \qquad \cot\theta = \frac{A}{O}$$

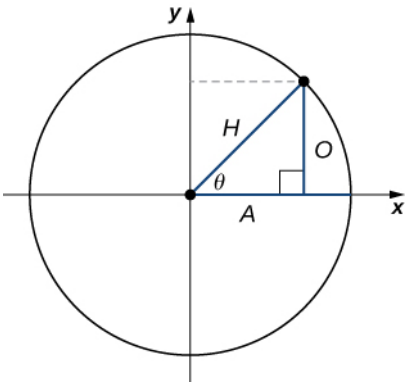


Figure 1.33 By inscribing a right triangle in a circle, we can express the ratios of the side lengths in terms of the trigonometric functions evaluated at θ.

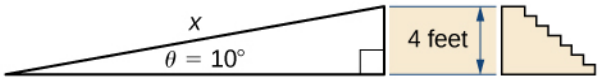
EXAMPLE 1.24

Constructing a Wooden Ramp

A wooden ramp is to be built with one end on the ground and the other end at the top of a short staircase. If the top of the staircase is 4 ft from the ground and the angle between the ground and the ramp is to be $10°$, how long does the ramp need to be?

Solution

Let x denote the length of the ramp. In the following image, we see that x needs to satisfy the equation $\sin(10°) = 4/x$. Solving this equation for x, we see that $x = 4/\sin(10°) \approx 23.035$ ft.



1.19 A house painter wants to lean a 20-ft ladder against a house. If the angle between the base of the ladder and the ground is to be $60°$, how far from the house should she place the base of the ladder?

Trigonometric Identities

A **trigonometric identity** is an equation involving trigonometric functions that is true for all angles θ for which the functions are defined. We can use the identities to help us solve or simplify equations. The main trigonometric identities are listed next.

Rule: Trigonometric Identities

Reciprocal identities

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$
$$\csc\theta = \frac{1}{\sin\theta} \qquad \sec\theta = \frac{1}{\cos\theta}$$

Pythagorean identities

$$\sin^2\theta + \cos^2\theta = 1 \qquad 1 + \tan^2\theta = \sec^2\theta \qquad 1 + \cot^2\theta = \csc^2\theta$$

Addition and subtraction formulas

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$
$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

Double-angle formulas

$$\sin(2\theta) = 2\sin\theta\cos\theta$$
$$\cos(2\theta) = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \cos^2\theta - \sin^2\theta$$

EXAMPLE 1.25

Solving Trigonometric Equations

For each of the following equations, use a trigonometric identity to find all solutions.

a. $1 + \cos(2\theta) = \cos\theta$
b. $\sin(2\theta) = \tan\theta$

Solution

a. Using the double-angle formula for $\cos(2\theta)$, we see that θ is a solution of

$$1 + \cos(2\theta) = \cos\theta$$

if and only if

$$1 + 2\cos^2\theta - 1 = \cos\theta,$$

which is true if and only if

$$2\cos^2\theta - \cos\theta = 0.$$

To solve this equation, it is important to note that we need to factor the left-hand side and not divide both sides of the equation by $\cos\theta$. The problem with dividing by $\cos\theta$ is that it is possible that $\cos\theta$ is zero. In fact, if we did divide both sides of the equation by $\cos\theta$, we would miss some of the solutions of the original equation. Factoring the left-hand side of the equation, we see that θ is a solution of this equation if and only if

$$\cos\theta\,(2\cos\theta - 1) = 0.$$

Since $\cos\theta = 0$ when

$$\theta = \frac{\pi}{2}, \frac{\pi}{2} \pm \pi, \frac{\pi}{2} \pm 2\pi, \ldots,$$

and $\cos\theta = 1/2$ when

$$\theta = \frac{\pi}{3}, \frac{\pi}{3} \pm 2\pi, \ldots \text{ or } \theta = -\frac{\pi}{3}, -\frac{\pi}{3} \pm 2\pi, \ldots,$$

we conclude that the set of solutions to this equation is

$$\theta = \frac{\pi}{2} + n\pi, \theta = \frac{\pi}{3} + 2n\pi, \text{and } \theta = -\frac{\pi}{3} + 2n\pi, n = 0, \pm 1, \pm 2, \ldots.$$

b. Using the double-angle formula for $\sin(2\theta)$ and the reciprocal identity for $\tan(\theta)$, the equation can be written as

$$2\sin\theta\cos\theta = \frac{\sin\theta}{\cos\theta}.$$

To solve this equation, we multiply both sides by $\cos\theta$ to eliminate the denominator, and say that if θ satisfies this equation, then θ satisfies the equation

$$2\sin\theta\cos^2\theta - \sin\theta = 0.$$

However, we need to be a little careful here. Even if θ satisfies this new equation, it may not satisfy the original equation because, to satisfy the original equation, we would need to be able to divide both sides of the equation by $\cos\theta$. However, if $\cos\theta = 0$, we cannot divide both sides of the equation by $\cos\theta$. Therefore, it is possible that we may arrive at extraneous solutions. So, at the end, it is important to check for extraneous solutions. Returning to the equation, it is important that we factor $\sin\theta$ out of both terms on the left-hand side instead of dividing both sides of the equation by $\sin\theta$. Factoring the left-hand side of the equation, we can rewrite this equation as

$$\sin\theta(2\cos^2\theta - 1) = 0.$$

Therefore, the solutions are given by the angles θ such that $\sin\theta = 0$ or $\cos^2\theta = 1/2$. The solutions of the first equation are $\theta = 0, \pm\pi, \pm 2\pi, \ldots$. The solutions of the second equation are $\theta = \pi/4, (\pi/4) \pm (\pi/2), (\pi/4) \pm \pi, \ldots$. After checking for extraneous solutions, the set of solutions to the equation is

$$\theta = n\pi \quad \text{and} \quad \theta = \frac{\pi}{4} + \frac{n\pi}{2}, n = 0, \pm 1, \pm 2, \ldots.$$

✓ 1.20 Find all solutions to the equation $\cos(2\theta) = \sin\theta$.

EXAMPLE 1.26

Proving a Trigonometric Identity

Prove the trigonometric identity $1 + \tan^2\theta = \sec^2\theta$.

Solution

We start with the identity

$$\sin^2\theta + \cos^2\theta = 1.$$

Dividing both sides of this equation by $\cos^2\theta$, we obtain

$$\frac{\sin^2\theta}{\cos^2\theta} + 1 = \frac{1}{\cos^2\theta}.$$

Since $\sin\theta/\cos\theta = \tan\theta$ and $1/\cos\theta = \sec\theta$, we conclude that

$$\tan^2\theta + 1 = \sec^2\theta.$$

1.21 Prove the trigonometric identity $1 + \cot^2\theta = \csc^2\theta$.

Graphs and Periods of the Trigonometric Functions

We have seen that as we travel around the unit circle, the values of the trigonometric functions repeat. We can see this pattern in the graphs of the functions. Let $P = (x, y)$ be a point on the unit circle and let θ be the corresponding angle . Since the angle θ and $\theta + 2\pi$ correspond to the same point P, the values of the trigonometric functions at θ and at $\theta + 2\pi$ are the same. Consequently, the trigonometric functions are **periodic functions.** The period of a function f is defined to be the smallest positive value p such that $f(x + p) = f(x)$ for all values x in the domain of f. The sine, cosine, secant, and cosecant functions have a period of 2π. Since the tangent and cotangent functions repeat on an interval of length π, their period is π (Figure 1.34).

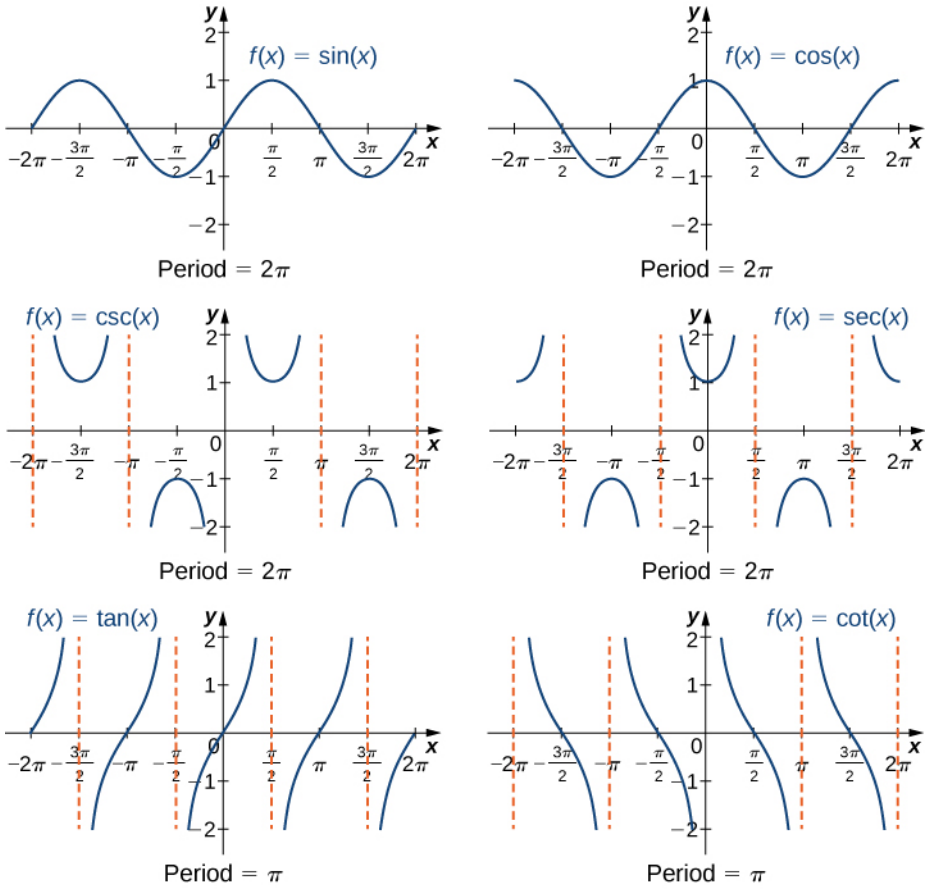


Figure 1.34 The six trigonometric functions are periodic.

Just as with algebraic functions, we can apply transformations to trigonometric functions. In particular, consider the following function:

$$f(x) = A\cos(B(x-\alpha)) + C. \tag{1.10}$$

In Figure 1.35, the constant α causes a horizontal or phase shift. The factor B changes the period. This transformed sine function will have a period $2\pi/|B|$. The factor A results in a vertical stretch by a factor of $|A|$. We say $|A|$ is the "amplitude of f." The constant C causes a vertical shift.

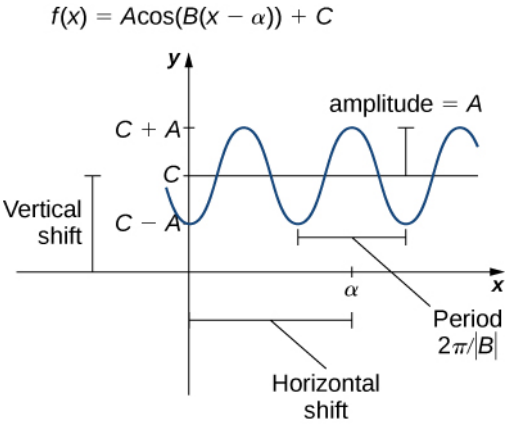


Figure 1.35 A graph of a general cosine function.

Notice in Figure 1.34 that the graph of $y = \cos x$ is the graph of $y = \sin x$ shifted to the left $\pi/2$ units. Therefore, we can write $\cos x = \sin(x + \pi/2)$. Similarly, we can view the graph of $y = \sin x$ as the graph of $y = \cos x$ shifted right $\pi/2$ units, and state that $\sin x = \cos(x - \pi/2)$.

A shifted sine curve arises naturally when graphing the number of hours of daylight in a given location as a function of the day of the year. For example, suppose a city reports that June 21 is the longest day of the year with 15.7 hours and December 21 is the shortest day of the year with 8.3 hours. It can be shown that the function

$$h(t) = 3.7\sin\left(\frac{2\pi}{365}(t - 80.5)\right) + 12$$

is a model for the number of hours of daylight h as a function of day of the year t (Figure 1.36).

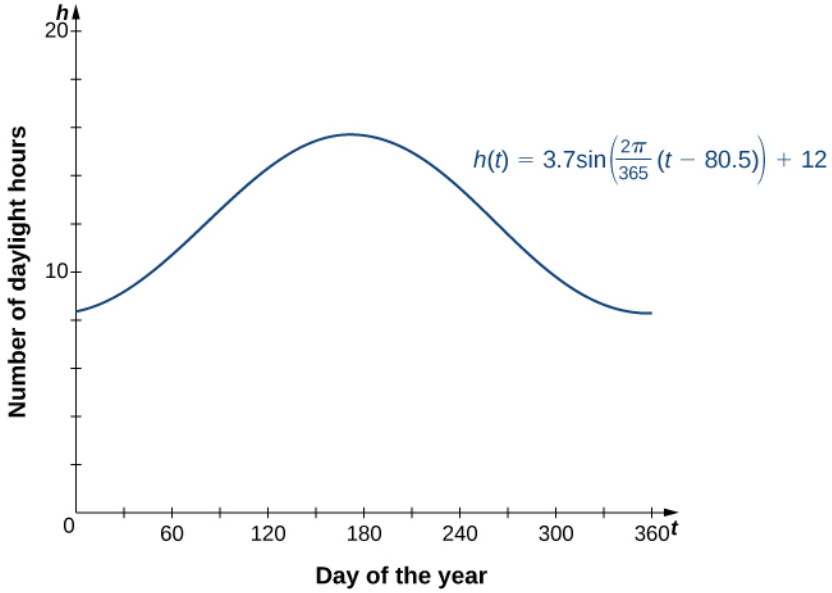


Figure 1.36 The hours of daylight as a function of day of the year can be modeled by a shifted sine curve.

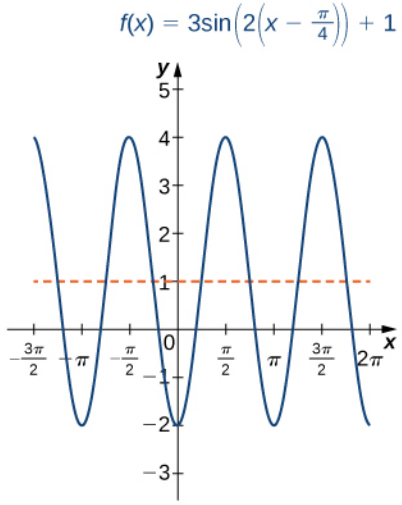
EXAMPLE 1.27

Sketching the Graph of a Transformed Sine Curve

Sketch a graph of $f(x) = 3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 1$.

Solution

This graph is a horizontal compression by a factor of 2, a phase shift to the right by π/4 units, followed by a vertical stretch by a factor of 3, and then a vertical shift by 1 unit. The period of f is π.



1.22 Describe the relationship between the graph of $f(x) = 3\sin(4x) - 5$ and the graph of $y = \sin(x)$.

SECTION 1.3 EXERCISES

For the following exercises, convert each angle in degrees to radians. Write the answer as a multiple of π.

113. $240°$

114. $15°$

115. $-60°$

116. $-225°$

117. $330°$

For the following exercises, convert each angle in radians to degrees.

118. $\frac{\pi}{2}$ rad

119. $\frac{7\pi}{6}$ rad

120. $\frac{11\pi}{2}$ rad

121. -3π rad

122. $\frac{5\pi}{12}$ rad

Evaluate the following functional values.

123. $\cos\left(\frac{4\pi}{3}\right)$

124. $\tan\left(\frac{19\pi}{4}\right)$

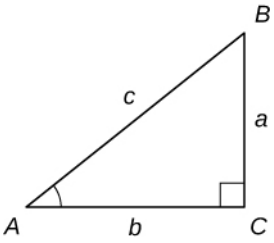
125. $\sin\left(-\frac{3\pi}{4}\right)$

126. $\sec\left(\frac{\pi}{6}\right)$

127. $\sin\left(\frac{\pi}{12}\right)$

128. $\cos\left(\frac{5\pi}{12}\right)$

For the following exercises, consider triangle ABC, a right triangle with a right angle at C. a. Find the missing side of the triangle. b. Find the six trigonometric function values for the angle at A. Where necessary, simplify to a fraction or round to three decimal places.



129. $a = 4, c = 7$

130. $a = 21, c = 29$

131. $a = 85.3, b = 125.5$

132. $b = 40, c = 41$

133. $a = 84, b = 13$

134. $b = 28, c = 35$

For the following exercises, P is a point on the unit circle. a. Find the (exact) missing coordinate value of each point and b. find the values of the six trigonometric functions for the angle θ with a terminal side that passes through point P. Rationalize denominators.

135. $P\left(\frac{7}{25}, y\right), y > 0$

136. $P\left(\frac{-15}{17}, y\right), y < 0$

137. $P\left(x, \frac{\sqrt{7}}{3}\right), x < 0$

138. $P\left(x, \frac{-\sqrt{15}}{4}\right), x > 0$

For the following exercises, simplify each expression by writing it in terms of sines and cosines, then simplify. The final answer does not have to be in terms of sine and cosine only.

139. $\tan^2 x + \sin x \csc x$

140. $\sec x \sin x \cot x$

141. $\frac{\tan^2 x}{\sec^2 x}$

142. $\sec x - \cos x$

143. $(1 + \tan\theta)^2 - 2\tan\theta$

144. $\sin x\,(\csc x - \sin x)$

145. $\frac{\cos t}{\sin t} + \frac{\sin t}{1+\cos t}$

146. $\frac{1+\tan^2\alpha}{1+\cot^2\alpha}$

For the following exercises, verify that each equation is an identity.

147. $\frac{\tan\theta\cot\theta}{\csc\theta} = \sin\theta$

148. $\frac{\sec^2\theta}{\tan\theta} = \sec\theta\csc\theta$

149. $\frac{\sin t}{\csc t} + \frac{\cos t}{\sec t} = 1$

150. $\frac{\sin x}{\cos x+1} + \frac{\cos x-1}{\sin x} = 0$

151. $\cot\gamma + \tan\gamma = \sec\gamma\csc\gamma$

152. $\sin^2\beta + \tan^2\beta + \cos^2\beta = \sec^2\beta$

153. $\frac{1}{1-\sin\alpha} + \frac{1}{1+\sin\alpha} = 2\sec^2\alpha$

154. $\frac{\tan\theta-\cot\theta}{\sin\theta\cos\theta} = \sec^2\theta - \csc^2\theta$

For the following exercises, solve the trigonometric equations on the interval $0 \le \theta < 2\pi$.

155. $2\sin\theta - 1 = 0$

156. $1 + \cos\theta = \frac{1}{2}$

157. $2\tan^2\theta = 2$

158. $4\sin^2\theta - 2 = 0$

159. $\sqrt{3}\cot\theta + 1 = 0$

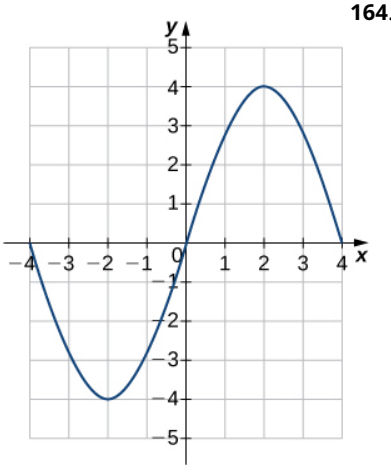
160. $3\sec\theta - 2\sqrt{3} = 0$

161. $2\cos\theta\sin\theta = \sin\theta$

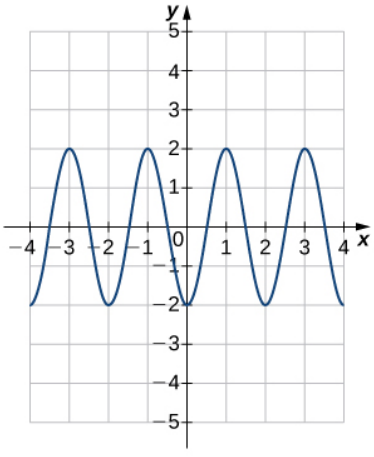
162. $\csc^2\theta + 2\csc\theta + 1 = 0$

For the following exercises, each graph is of the form $y = A\sin Bx$ or $y = A\cos Bx$, where $B > 0$. Write the equation of the graph.

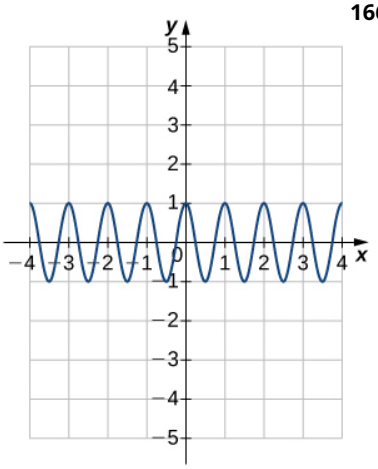
163.



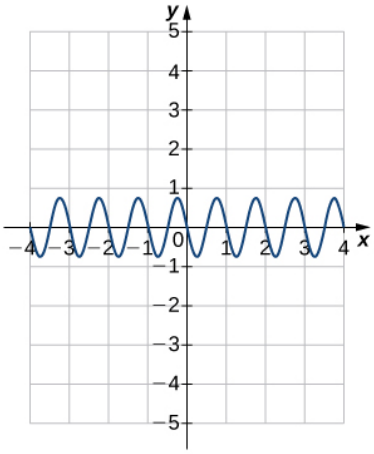
164.



165.



166.



For the following exercises, find a. the amplitude, b. the period, and c. the phase shift with direction for each function.

167. $y = \sin\left(x - \frac{\pi}{4}\right)$

168. $y = 3\cos(2x + 3)$

169. $y = \frac{-1}{2}\sin\left(\frac{1}{4}x\right)$

170. $y = 2\cos\left(x - \frac{\pi}{3}\right)$

171. $y = -3\sin(\pi x + 2)$

172. $y = 4\cos\left(2x - \frac{\pi}{2}\right)$

173. **[T]** The diameter of a wheel rolling on the ground is 40 in. If the wheel rotates through an angle of $120°$, how many inches does it move? Approximate to the nearest whole inch.

174. **[T]** Find the length of the arc intercepted by central angle θ in a circle of radius r. Round to the nearest hundredth.
a. $r = 12.8$ cm, $\theta = \frac{5\pi}{6}$ rad b. $r = 4.378$ cm, $\theta = \frac{7\pi}{6}$ rad c. $r = 0.964$ cm, $\theta = 50°$ d. $r = 8.55$ cm, $\theta = 325°$

175. **[T]** As a point P moves around a circle, the measure of the angle changes. The measure of how fast the angle is changing is called *angular speed*, ω, and is given by $\omega = \theta/t$, where θ is in radians and t is time. Find the angular speed for the given data. Round to the nearest thousandth.
a. $\theta = \frac{7\pi}{4}\text{rad}, t = 10$ sec b. $\theta = \frac{3\pi}{5}\text{rad}, t = 8$ sec c. $\theta = \frac{2\pi}{9}\text{rad}, t = 1$ min d. $\theta = 23.76\,\text{rad}, t = 14$ min

176. **[T]** A total of 250,000 m^2 of land is needed to build a nuclear power plant. Suppose it is decided that the area on which the power plant is to be built should be circular.
a. Find the radius of the circular land area.
b. If the land area is to form a $45°$ sector of a circle instead of a whole circle, find the length of the curved side.

177. **[T]** The area of an isosceles triangle with equal sides of length x is $\frac{1}{2}x^2\sin\theta$, where θ is the angle formed by the two sides. Find the area of an isosceles triangle with equal sides of length 8 in. and angle $\theta = 5\pi/12$ rad.

178. **[T]** A particle travels in a circular path at a constant angular speed ω. The angular speed is modeled by the function $\omega = 9\,|\cos(\pi t - \pi/12)|$. Determine the angular speed at $t = 9$ sec.